

Midterm Reference Sheet

SN1 Sections 2 and 4, Fall 2025

Basic Operations

```
# addition, subtraction, division, multiplication, exponentiation
result = 19 + 17 # result gets the value 26
result = 19 - 17 # result gets the value 2
result = 19 / 17 # result gets the value 1.1176470588235294
result = 4 / 2   # result gets the value 2.0
result = 19 * 17 # result gets the value 323
result = 2 ** 4  # result gets the value 16

# integer division -- if you provide two integers, you get an integer
result
result = 19 // 17 # result gets the value 1
result = 4 // 2   # result gets the value 2

# modulus -- returns the remainder after division
result = 19 % 17 # result gets the value 2
result = 17 % 19 # result gets the value 17
result = 19 % 2  # result gets the value 1
result = 4 % 2   # result gets the value 0
```

Boolean Operations

```
# equal to '==', not equal to '!='
result = 19 == 17 # result gets the value True
result = 19 != 17 # result gets the value False

# less than '<', greater than '>'
result = 19 > 17 # result gets the value True
result = 19 < 17 # result gets the value False
result = 19 < 19 # result gets the value False

# less than or equal to '<=', greater than or equal to '>='
result = 19 >= 17 # result gets the value True
result = 19 <= 17 # result gets the value False
result = 19 <- 19 # result gets the value True
```

```
# combining booleans using 'and'
result = 19 == 17 and 19 < 17 # result gets the value False
result = 19 == 17 and 19 > 17 # result gets the value True

# combining booleans using 'or'
result = 19 == 17 or 19 > 17 # result gets the value True
result = 19 == 17 or 19 < 17 # result gets the value True
result = 19 != 17 or 19 < 17 # result gets the value False
```

Useful standard functions

```
# result gets the value of the input as a string
result = input("Enter a string value: ")

# result gets the value of the input converted to an int
result = int(input("Enter an integer value: "))

# result gets the value of the input converted to a float
result = float(input("Enter any number: "))
```

Useful math functions

```
# Don't forget to import the math library if you're going to use it!
import math

result = math.ceil(6.1) # result gets the value 7
result = math.floor(6.1) # result gets the value 6
result = abs(-7) # result gets the value 7

# trigonometry
theta = math.pi # theta gets the value pi (remember pi==180
degrees)
result = math.sin(theta) # result gets the value 0.0
result = math.tan(theta) # result gets the value 0.0
result = math.cos(theta) # result gets the value -1.0
```

Useful turtle functions

```
# Don't forget to import the turtle library if you're going to use it!
import turtle

# Turtle movement
turtle.forward(100)
turtle.backward(50)
```

```

turtle.goto(100, 100)
turtle.teleport(200, 200)

# Turn the default turtle (angle given in degrees)
turtle.right(30)
turtle.left(50)
turtle.setheading(0) # 0 is east, 90 is north, etc.

# Turtle drawing
turtle.up() # lift turtle pen so no path drawing takes place
turtle.down() # put turtle pen back down

turtle.pencolor("blue") # change turtle path color to blue
turtle.pensize(3) # sets the pen width to 3 pixels
turtle.fillcolor("red") # change fill color to red

turtle.begin_fill() # turtle will fill in space between lines
turtle.end_fill() # turtle will stop filling in space between lines

# Create a new turtle
new_turtle = turtle.Turtle()
# Use the new turtle
new_turtle.forward(100)
# Clone an existing turtle
turtle_clone = new_turtle.clone()

# Ends the turtle program
turtle.mainloop()

```

Conditional Logic

```

# let condition_1 and condition_2 be two different boolean variables:
if condition_1:
    # this code will only execute if condition_1 is True
    result = "condition_1 True"
elif condition_2:
    # this code will only execute if condition_1 is False and condition_2
    is True
    result = "condition_2 True"
else:
    # this code will only execute if both conditions above are False
    result = "neither condition_1 nor condition_2 True"

print(result) # This code will always execute no matter what conditions
are true

```

Looping

```
# i goes from 0 .. 9
for i in range(10):
# i goes from 3 .. 9
for i in range(3, 10):
# i gets the value 3, then 5, then 7, then 9
for i in range(3, 10, 2):
# fruit gets the value "apple", then "banana", then "orange"
for fruit in ["apple", "banana", "orange"]:
```

Custom Functions

```
# An example function definition:
def foo(a, b):      # function 'foo' takes two parameters 'a' and 'b'
    return a + b    # returns the result of adding 'a' and 'b'

# Using the example function:
result = foo(1, 2)   # result gets the value 3
result = foo("1", "2") # result gets the value "12"
```